

5. inclusion-exclusion identity

$$4. P(E_1 \cup E_2 \cup E_3)$$

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = \sum_{i=1}^n P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} E_{i_2})$$

$$+ \dots + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r})$$

$$+ \dots + (-1)^{n+1} P(E_1 E_2 \dots E_n)$$

$$\text{proof: } P(E_1 \cup E_2 \cup E_3)$$

$$\text{i.e., } P\left(\bigcup_{i=1}^n E_i\right) = \sum_{r=1}^n (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r})$$

$$= P((E_1 \cup E_2) \cup E_3)$$

$$= P(E_1) + P(E_2) - P(E_1 E_2)$$

5.3 proof: (method 1)

$$\text{here } E_2 = E_1 \cup E_2$$

$$\therefore P(E_2) = P(E_1) + P(E_2) - P(E_1 E_2)$$

$$P(E_1 E_2) = P(E_1 E_2 \cup E_3)$$

$$= P(E_1 E_2) + P(E_1 E_3)$$

$$- P(E_1 E_2 E_3)$$

$$\therefore P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$$

$$- P(E_1 E_2) - P(E_1 E_3) - P(E_2 E_3)$$

$$+ P(E_1 E_2 E_3)$$

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$$= P\left(\bigcup_{i=1}^n E_i\right) + P(E_n) - P\left(\bigcup_{i=1}^n E_i\right) E_n$$

$$\therefore P\left(\bigcup_{i=1}^n E_i\right) = \sum_{r=1}^n (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r})$$

inclusion-exclusion identity

$$1. P(E^c) = 1 - P(E)$$

$$\text{proof: } 1 = P(S) = P(E \cup E^c) = P(E) + P(E^c)$$

$$\therefore P(E^c) = 1 - P(E)$$

$$2. E \subset F \Rightarrow P(E) \leq P(F)$$

$$\text{proof: } \because F = E \cup (F \setminus E) = E \cup (E^c F)$$

$$\therefore P(F) = P(E) + P(E^c F)$$

$$\therefore P(F) \geq P(E)$$

$$3. P(E \cup F) = P(E) + P(F) - P(EF)$$

$$\text{proof: } \because E \cup F = E \cup E^c F$$

$$\therefore P(E \cup F) = P(E) + P(E^c F)$$

$$\because F = EF \cup E^c F$$

$$\therefore P(F) = P(EF) + P(E^c F)$$

$$\therefore P(E^c F) = P(F) - P(EF)$$

$$\therefore P(E \cup F) = P(E) + P(F) - P(EF)$$

54 method 2, see 10.

$$= (\bigcup_{i=1}^n E_i) \cup (E_1^c \cap \dots \cap E_n^c) E_n$$

$$= E_1 \cup E_2 \cup \dots \cup E_n \cup E_1^c \cap E_2^c \cap \dots \cap E_n^c \cup E_n$$

$$\therefore l=r$$

□

proof: ① $n=1, l=r$.

$$7. P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) - \sum_{j < k} P(E_j \cap E_k) + \dots$$

$$n=2, l = E_1 \cup E_2$$

$$= E_1 \cup (E_2 \setminus E_1)$$

$$\text{proof: ① } \bigcup_{i=1}^n E_i = E_1 \cup E_2 \cup \dots \cup E_n$$

$$= E_1 \cup E_2^c \cap E_2 = E_1 \cup E_2$$

$\therefore$ right is union of disjoint events

$$n=3, l = E_1 \cup E_2 \cup E_3$$

$$\therefore P(\bigcup_{i=1}^n E_i) = P(E_1) + \sum_{i=2}^n P(E_i^c \cap E_1)$$

$$\text{② } \forall B, P(E) = P(B \cap E) + P(B^c \cap E)$$

$$\therefore P(E_1) = P(E_1^c \cap E_1) + P(E_1 \cap E_1)$$

$$= P(E_1^c \cap E_1) + P(\bigcup_{j=1}^{i-1} E_j \cap E_1)$$

$$= P(E_1^c \cap E_1) + P(\bigcup_{j < i} E_j \cap E_1)$$

$$\therefore P(E_1^c \cap E_1) = P(E_1) - P(\bigcup_{j=1}^{i-1} E_j \cap E_1)$$

$$\text{③ } \therefore P(\bigcup_{i=1}^n E_i) = P(E_1) + \sum_{i=2}^n P(E_i) - \sum_{j < k} P(E_j \cap E_k) + \dots$$

$$\therefore P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) - \sum_{j < k} P(E_j \cap E_k) + \dots$$

$$\therefore P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) - \sum_{j < k} P(E_j \cap E_k) + \dots$$

$$P(\bigcup_{i=1}^n E_i | E_n) = P(\bigcup_{i=1}^n (E_i \cap E_n))$$

$$= \sum_{i=1}^{n-1} (-1)^{i+1} \sum_{i_1 < i_2 < \dots < i_r < n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r} \cap E_n)$$

$$\therefore P(E_n) - \sum_{i=1}^{n-1} (-1)^{i+1} \sum_{i_1 < i_2 < \dots < i_r < n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r} \cap E_n)$$

$$= P(E_n) + \sum_{i=1}^{n-1} (-1)^{i+1} \sum_{i_1 < i_2 < \dots < i_r < n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r} \cap E_n)$$

$$= \sum_{i=1}^n (-1)^{i+1} \sum_{i_1 < i_2 < \dots < i_r = n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^{n-1} (-1)^{i+1} \sum_{i_1 < i_2 < \dots < i_r < n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$+ \sum_{i=1}^n (-1)^{i+1} \sum_{i_1 < i_2 < \dots < i_r = n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

$$= \sum_{i=1}^n (-1)^{i+1} \sum_{i_1 < i_2 < \dots < i_r = n} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

□

8.3 $P(\bigcup_{i=1}^n E_i)$

$$\leq \sum_{i=1}^n P(E_i) - \sum_{i,j: i < j} P(E_i E_j)$$

$$+ \sum_{i,j,k: i < j < k} P(E_i E_j E_k)$$

proof: ① use 8.2,

$$\forall i \in \mathbb{Z}, P(\bigcup_{j: j < i} E_j) \geq \sum_{j: j < i} P(E_j)$$

$$- \sum_{k: j < k < i} P(E_j E_k E_i)$$

$$\textcircled{2} \text{ from 8.0, } P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) - \sum_{i=1}^n \sum_{j: j < i} P(E_i E_j)$$

$$\leq \sum_{i=1}^n P(E_i) - \sum_{i=1}^n \left(\sum_{j: j < i} P(E_j) - \sum_{k: j < k < i} P(E_j E_k E_i) \right)$$

$$= \sum_{i=1}^n P(E_i) - \sum_{i,j: i < j} P(E_i E_j) + \sum_{i,j,k: i < j < k} P(E_i E_j E_k)$$

$n \in \mathbb{Z}$

$$(\text{note } \sum_{i,j,k: i < j < k} P(E_i E_j E_k) = 0 \text{ if } n < 3)$$

$$\textcircled{3} \text{ if } n \leq 0, \bigcup_{i=1}^n E_i := \emptyset$$

$$\sum_{i=1}^n P(E_i) := 0$$

$$+ \sum_{i,j,k: i < j < k} P(E_i E_j E_k), n \in \mathbb{Z} \quad \therefore P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) = 0$$

$$8.2 \quad P(\bigcup_{i=1}^n E_i) \geq \sum_{i=1}^n P(E_i) - \sum_{i,j: i < j} P(E_i E_j)$$

proof: ① use 8.1,

$$P(\bigcup_{i=1}^n E_i) \leq \sum_{j: j < i} P(E_i E_j), \forall i \in \mathbb{Z}$$

from 8.0,

$$P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) - \sum_{i=1}^n \sum_{j: j < i} P(E_i E_j)$$

$$\geq \sum_{i=1}^n P(E_i) - \sum_{i=1}^n \sum_{j: j < i} P(E_j)$$

$$= \sum_{i=1}^n P(E_i) - \sum_{i,j: i < j} P(E_i E_j)$$

$$(\text{note } \sum_{i,j: i < j} P(E_i E_j) = 0, \text{ if } n < 2)$$

8.0

$$P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) - \sum_{i=2}^n \sum_{j: j < i} P(E_i E_j)$$

$$= \sum_{i=1}^n P(E_i) - \sum_{i=2}^n P(\bigcup_{j=1}^{i-1} E_j E_i)$$

$$= \sum_{i=1}^n P(E_i) - \sum_{i=1}^n P(\bigcup_{j=1}^{i-1} E_j E_i)$$

$$(\text{here } \bigcup_{j=1}^{i-1} E_j E_i = \emptyset \text{ if } i=1)$$

$$= \sum_{i=1}^n P(E_i) - \sum_{i=1}^n P(\bigcup_{j: j < i} E_j E_i), n \in \mathbb{Z}$$

(if $n \leq 0, n=0$)

$$P(\bigcup_{i=1}^n E_i) \leq \sum_{i=1}^n P(E_i), n \in \mathbb{Z}$$

proof: ① $\therefore \sum_{i=1}^n P(\bigcup_{j: j < i} E_j E_i) \geq 0$

$$\textcircled{2} \text{ from 8.1, } P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i) - \sum_{i=1}^n \sum_{j: j < i} P(E_i E_j)$$

$$\therefore P(\bigcup_{i=1}^n E_i) \leq \sum_{i=1}^n P(E_i), n \geq 1$$

$$= \sum_{i=1}^n P(E_i) - \sum_{i=1}^n \sum_{r=2}^j (-1)^{r+1} \sum_{i_1, i_2, \dots, i_r: i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r}) \quad (7)$$

$$= \sum_{i=1}^n P(E_i) + \sum_{r=2}^{j+1} (-1)^{r+1} \sum_{i_1, i_2, \dots, i_r: i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r})$$

$$= \sum_{i=1}^{j+1} (-1)^{r+1} \sum_{i_1, i_2, \dots, i_r: i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r}) \quad \text{then } \forall i \in \mathbb{Z},$$

$\therefore$ for $k=j+1$, 9.1 hold.

$\therefore \forall k \in \mathbb{Z}^+$, 9.1 hold. $\square$

$$10. P(\bigcup_{i=1}^n E_i) = \sum_{i=1}^n (-1)^{r+1} \sum_{i_1, i_2, \dots, i_r: i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r}) \quad (8)$$

proof: (method 2 of 5)

for 9.1, $\because k \in \mathbb{Z}^+$

$\therefore$ let $k=n \Rightarrow 1 \leq \dots \leq n$ from 8.10

let $k=n+1 \Rightarrow 1 \leq \dots \leq n+1$

$\square$

$$k=j \Rightarrow P(\bigcup_{i=1}^j E_i)$$

$$\left(\sum_{i=1}^{j+1} (-1)^{r+1} \sum_{i_1, i_2, \dots, i_r: i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r}) \right) \quad \forall n \in \mathbb{Z}$$

then $\forall i \in \mathbb{Z}$,

$$P\left(\bigcup_{i_2: i_2 < i_1} E_{i_1} E_{i_2}\right) = P\left(\bigcup_{i_2=1}^{i_1-1} E_{i_1} E_{i_2}\right)$$

$$= P\left(\bigcup_{h=1}^{i_1-1} E_{i_1} E_h\right) = P\left(\bigcup_{h=1}^{i_1-1} F_h\right), \quad (F_h = E_{i_1} E_h)$$

$$\sum_{r=1}^{j+1} (-1)^{r+1} \sum_{h_1, h_2, \dots, h_r: h_1 < h_2 < \dots < h_r} P(F_{h_1} F_{h_2} \dots F_{h_r})$$

$$= \sum_{r=1}^{j+1} (-1)^{r+1} \sum_{h_1, h_2, \dots, h_r: h_1 < h_2 < \dots < h_r} P(E_{i_1} E_{h_1} E_{h_2} \dots E_{h_r})$$

$$= \sum_{r=1}^{j+1} (-1)^{r+1} \sum_{i_2, i_3, \dots, i_{r+1}: i_1 < i_2 < i_3 < \dots < i_{r+1}} P(E_{i_1} E_{i_2} E_{i_3} \dots E_{i_{r+1}})$$

$\therefore$ from 8.10

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_{i_1}) - \sum_{i=1}^n \sum_{i_2: i_2 < i_1} P\left(\bigcup_{i_2=1}^{i_1-1} E_{i_1} E_{i_2}\right)$$

$$\left(\sum_{i=1}^{j+2} P(E_{i_1}) - \sum_{i=1}^n \sum_{i_2: i_2 < i_1} (-1)^{r+1} \sum_{i_3, i_4, \dots, i_{r+1}: i_1 < i_2 < i_3 < \dots < i_{r+1}} P(E_{i_1} E_{i_2} \dots E_{i_{r+1}}) \right)$$

$$9.1 \quad P\left(\bigcup_{i=1}^k E_i\right) \quad (9)$$

$$\sum_{r=1}^k (-1)^{r+1} \sum_{i_1, i_2, \dots, i_r: i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r}) \quad k \in \mathbb{Z}^+$$

here $\left(\sum_{i=1}^k \right)^k : \begin{cases} \leq, & \text{if } k \text{ even} \\ \geq, & \text{if } k \text{ odd} \end{cases}$

$\left(\sum_{i=1}^k \right)^k : \begin{cases} \geq, & \text{if } k \text{ even} \\ \leq, & \text{if } k \text{ odd} \end{cases}$

9.2 proof: (by induction)

let $k=1 \Rightarrow P\left(\bigcup_{i=1}^1 E_i\right) \leq \sum_{i=1}^1 P(E_i), \quad \forall n \in \mathbb{Z}^+$

let $k=2 \Rightarrow P\left(\bigcup_{i=1}^2 E_i\right) \geq \sum_{i=1}^2 P(E_i) - \sum_{i_1, i_2: i_1 < i_2} P(E_{i_1} E_{i_2})$

let $k=3 \Rightarrow P\left(\bigcup_{i=1}^3 E_i\right) \leq \sum_{i=1}^3 P(E_i) - \sum_{i_1, i_2: i_1 < i_2} P(E_{i_1} E_{i_2}) + \sum_{i_1, i_2, i_3: i_1 < i_2 < i_3} P(E_{i_1} E_{i_2} E_{i_3}), \quad \forall n \in \mathbb{Z}$

$\square$

$$0.1 \quad \forall E, F,$$

$$\Rightarrow F = EF \cup E^c F$$

$$\begin{aligned} \text{proof: } F &= SF = (E \cup E^c)F \\ &= EF \cup E^c F \end{aligned}$$

$$0.2 \quad \forall E, F$$

$$\Rightarrow E \cup F = E \cup E^c F$$

$$\text{proof: } \because E \cup F = E \cup (EF \cup E^c F)$$

$$\begin{aligned} \therefore E \cup F &= (E \cup EF) \cup E^c F \\ &= E \cup E^c F \end{aligned}$$

$$0.3 \quad \forall E, F, E \subset F$$

$$\Rightarrow F = E \cup E^c F$$

$$\text{proof: } \because F = E \cup F$$

$$\therefore F = E \cup E^c F$$